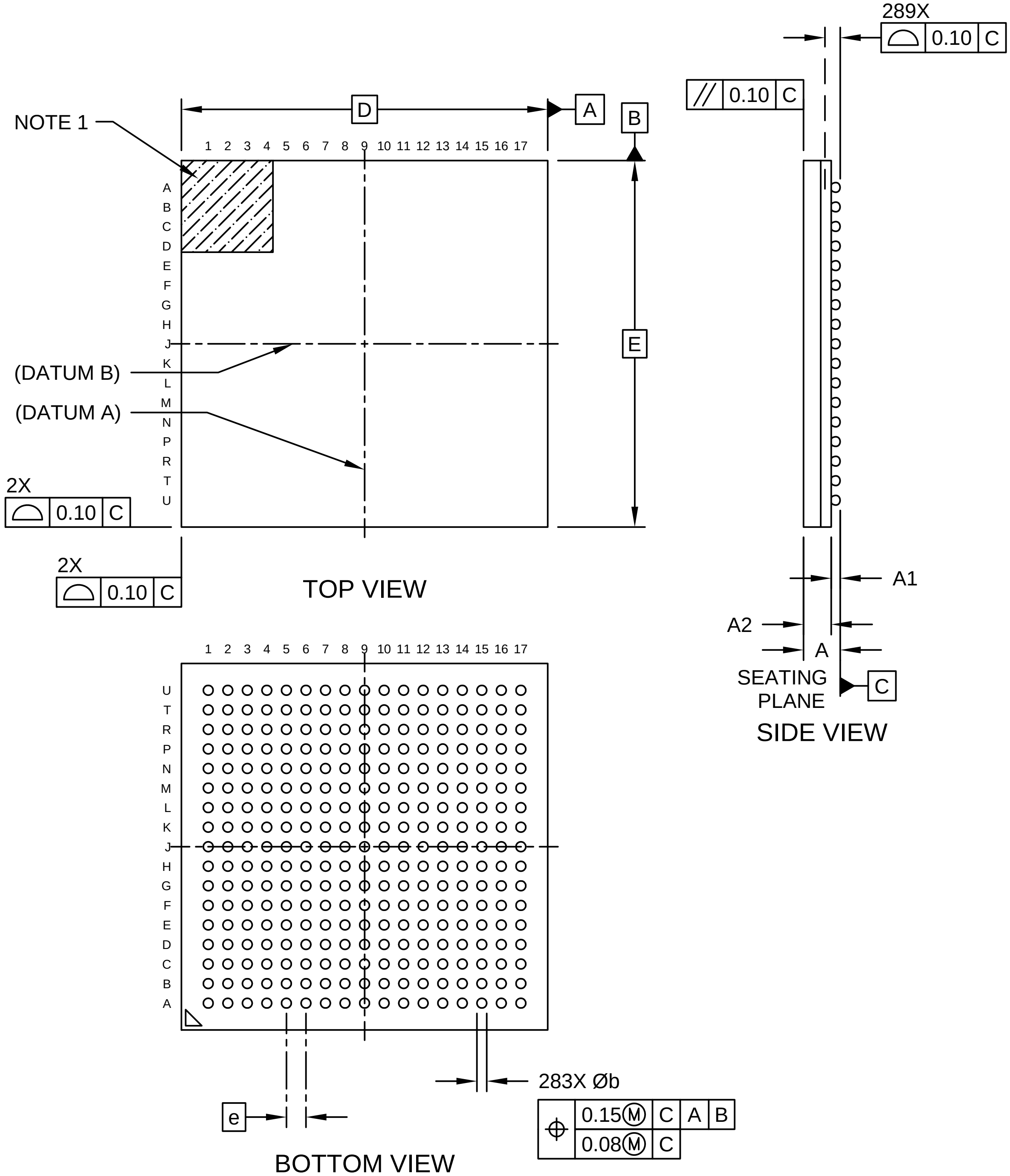
289-Ball Low Profile fine Pitch Ball Grid Array (AQB) - 15x15x1.46 mm Body [LFBGA] Atmel LegacyGlobal Package Code CEG

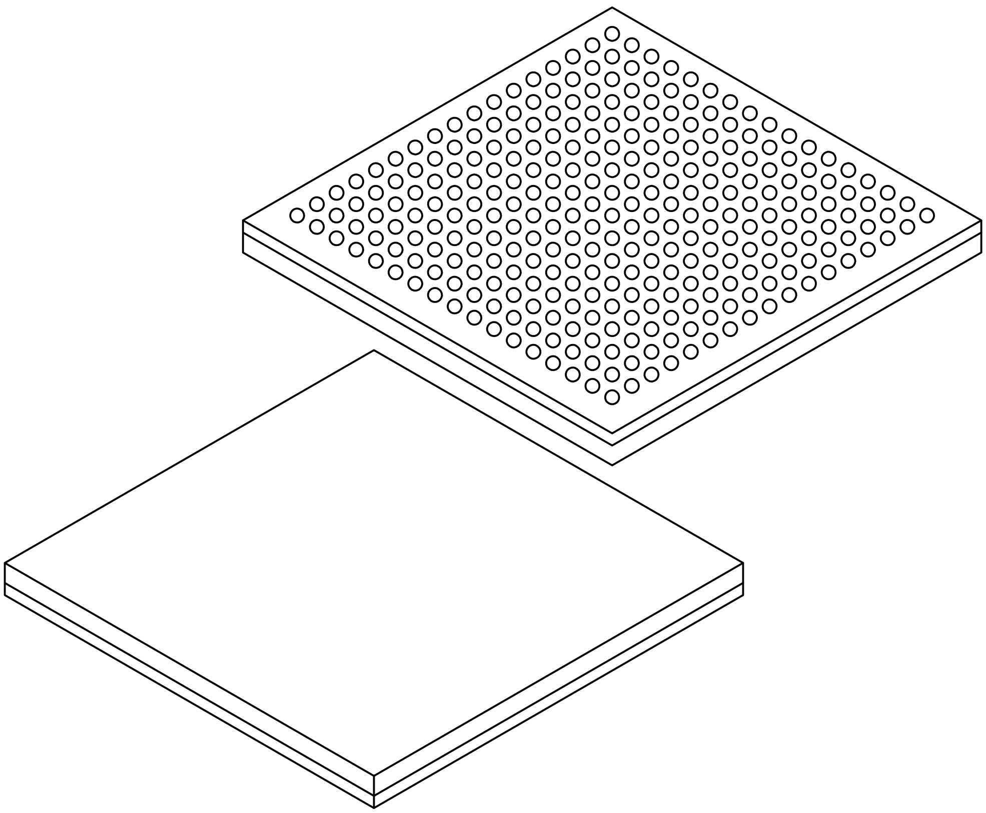
Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Microchip Technology Drawing C04-21127 Rev A Sheet 1 of 2

289-Ball Low Profile fine Pitch Ball Grid Array (AQB) - 15x15x1.50 mm Body [LFBGA] Atmel LegacyGlobal Package Code CEG

Note: For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



	Units	MILLIMETERS		
	Dimension Limits	MIN	NOM	MAX
Number of Terminals	N	289		
Pitch	e	0.80 BSC		
Overall Height	A	1.30	1.40	1.50
Standoff	A1	0.32	0.37	0.42
Terminal Thickness	A2	0.65	0.70	0.75
Overall Length	D	15.00 BSC		
Overall Width	E	15.00 BSC		
Terminal Width	b	0.35	0.40	0.45

Notes:

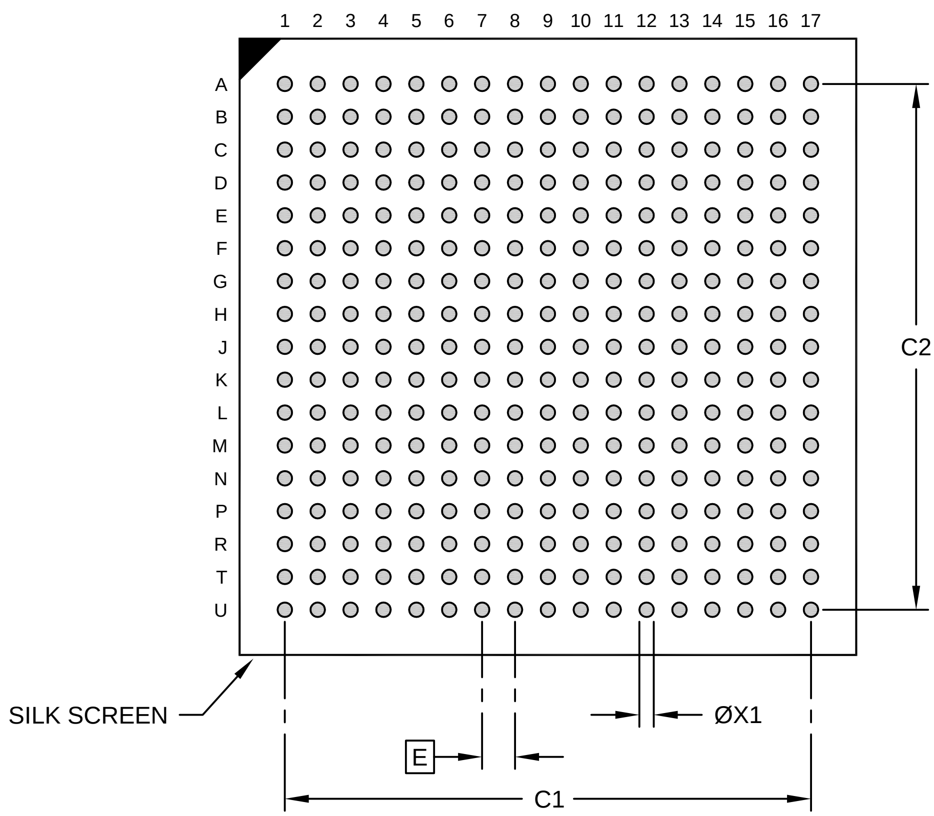
1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M

   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

   REF: Reference Dimension, usually without tolerance, for information purposes only.

289-Ball Low Profile fine Pitch Ball Grid Array (AQB) - 15x15x1.46 mm Body [LFBGA] Atmel LegacyGlobal Package Code CEG

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

		Units	MILLIMETERS	
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1		12.80	
Contact Pad Spacing	C2		12.80	
Contact Pad Width (X20)	X1			0.35

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23127 Rev A